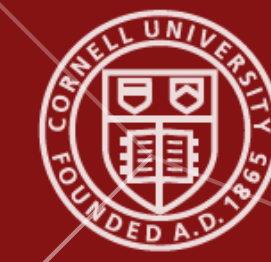


Temporal Abstraction in World Modeling for Long-Horizon Event Prediction

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Visual overview

Long-horizon prediction needs a world model that decides how far ahead to reason. The Predictron blends predictions over many imagined depths, and the mixture weights w^k segment a trajectory into the spans each depth accounts for, so the horizon is learned rather than fixed.

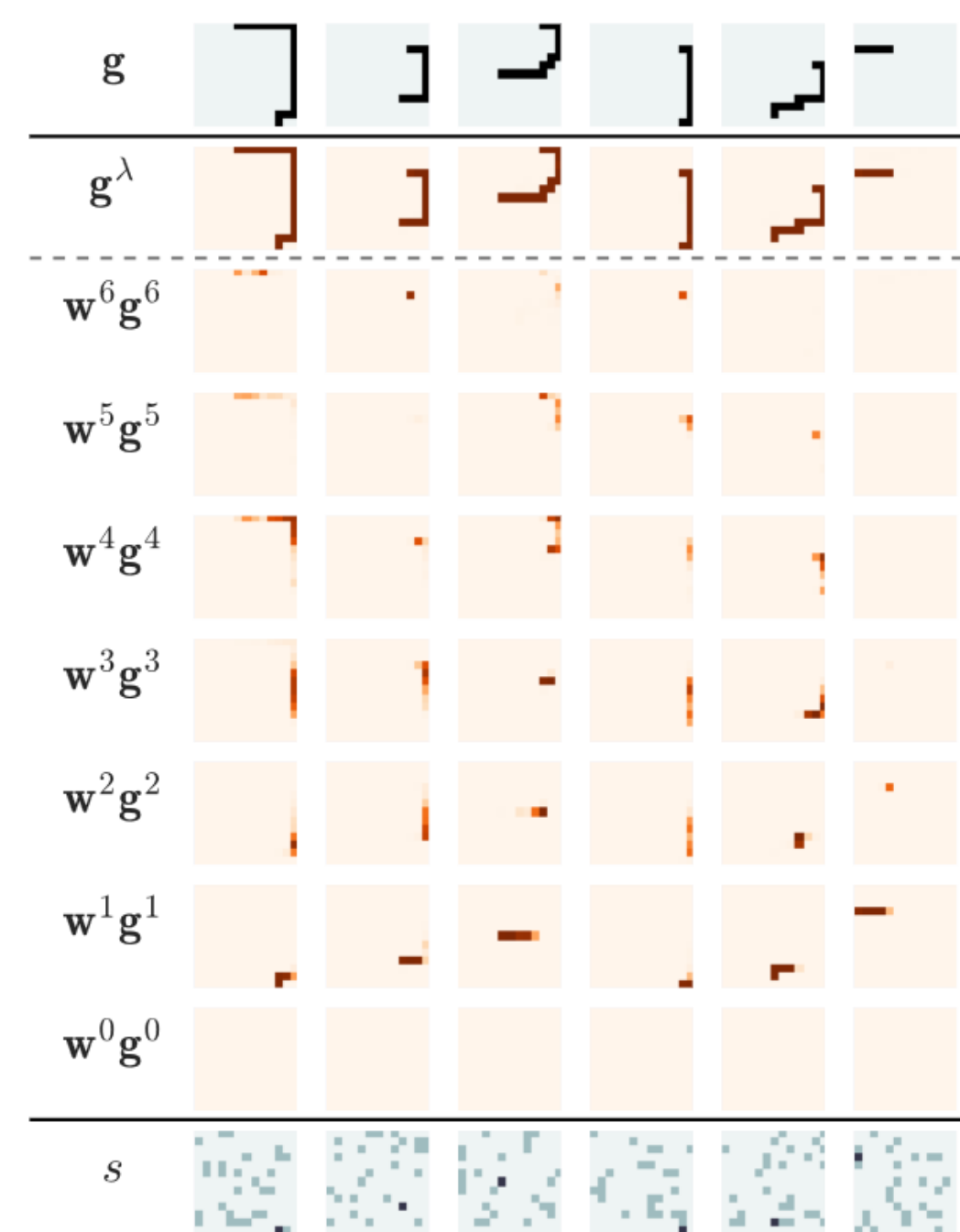
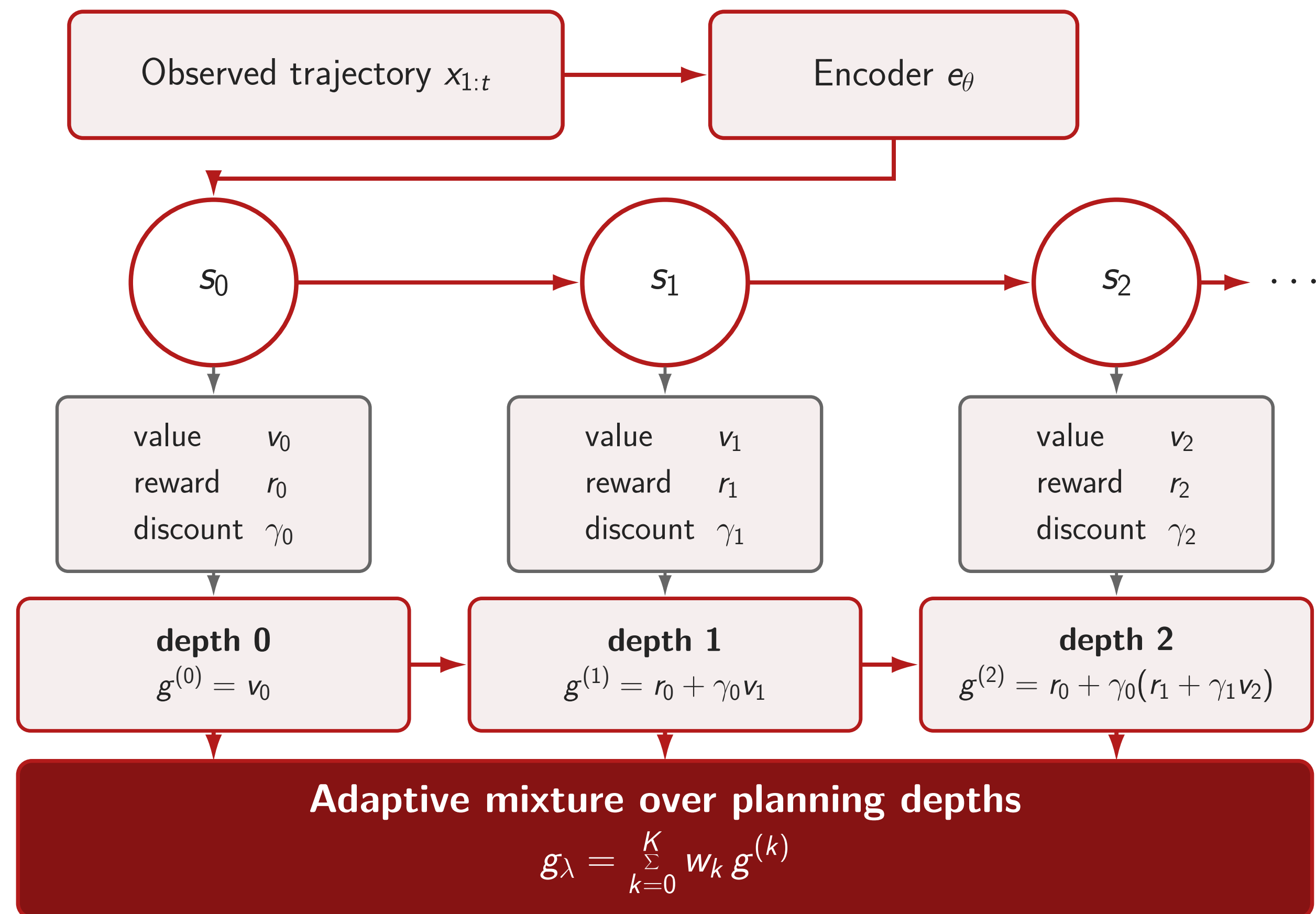


Figure 1: Predictron architecture [1] (left); the λ -return g^λ against ground truth g , split by planning depth (right).

Predictron

Loss function

$$g^\lambda = \sum_{k=0}^K w_k g^k \quad (1)$$

$$w^k = \begin{cases} (1 - \lambda^k) \prod_{j=0}^{k-1} \lambda^j & \text{if } k < K \\ \prod_{j=0}^{K-1} \lambda^j & \text{if } k = K \end{cases} \quad (2)$$

$$L^{0:K} = \frac{1}{2K} \sum_{k=0}^K \|\mathbb{E}_p[g | s] - \mathbb{E}_m[g^k | s]\|^2 \quad (3)$$

$$L^\lambda = \frac{1}{2} \|\mathbb{E}_p[g | s] - \mathbb{E}_m[g^\lambda | s]\|^2 \quad (4)$$

JESA: joint embedding successor architecture

JESA regresses the predictron's own latent onto a *separate* base encoder ϕ , never onto pixels. The hatch marks the cut gradient, which stops both branches collapsing to a constant.

$$\mathcal{L}_{\text{JESA}} = \mathbb{E} \left\| \phi(s_0) - \sum_{t=0}^{\infty} \gamma^t \phi(s_{t+1}) \right\|^2$$

Reward is linear in ϕ , so Q is linear in ψ : one weight vector per action, fitted linearly.

$$Q(s, a) = \mathbf{w}_a^\top \psi(s)$$

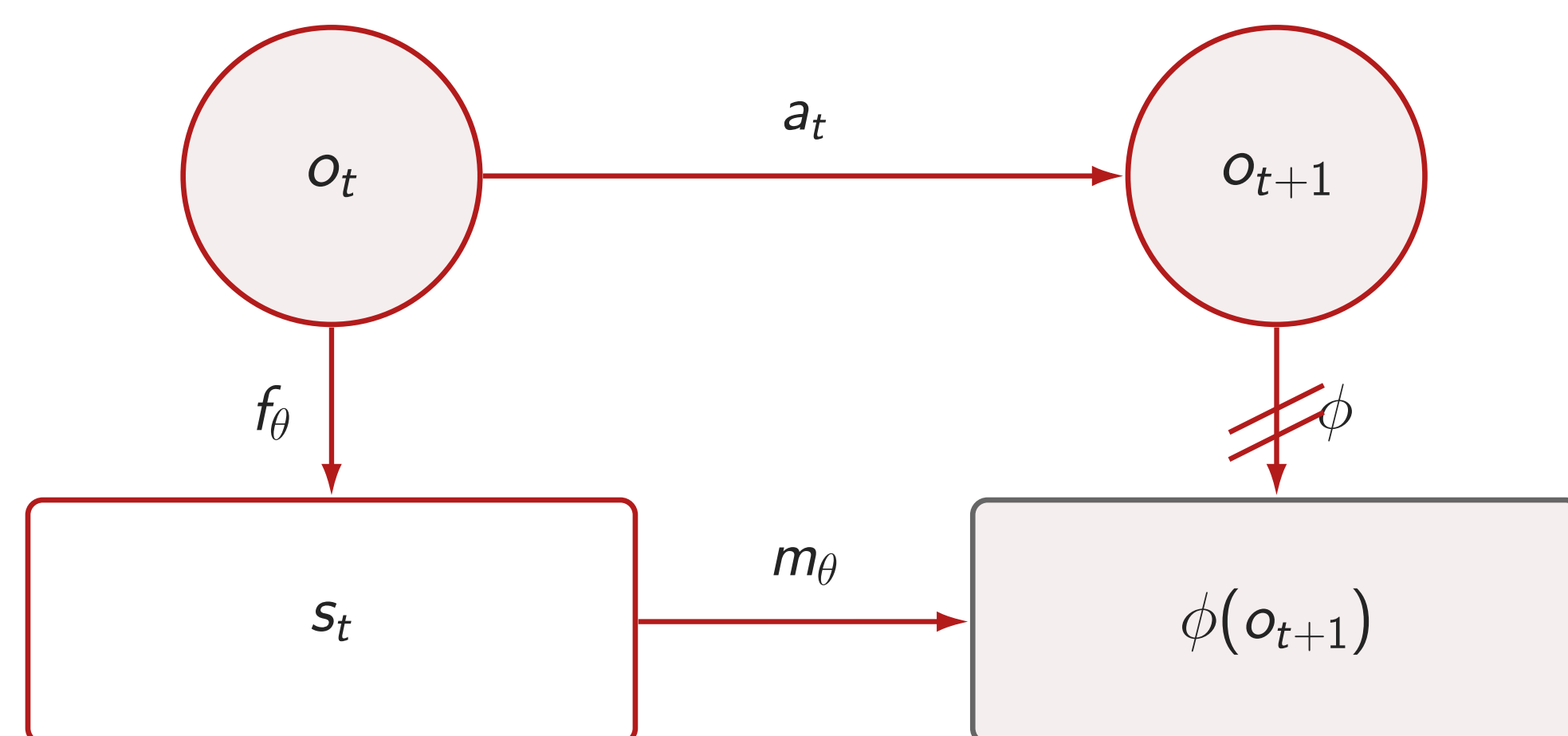
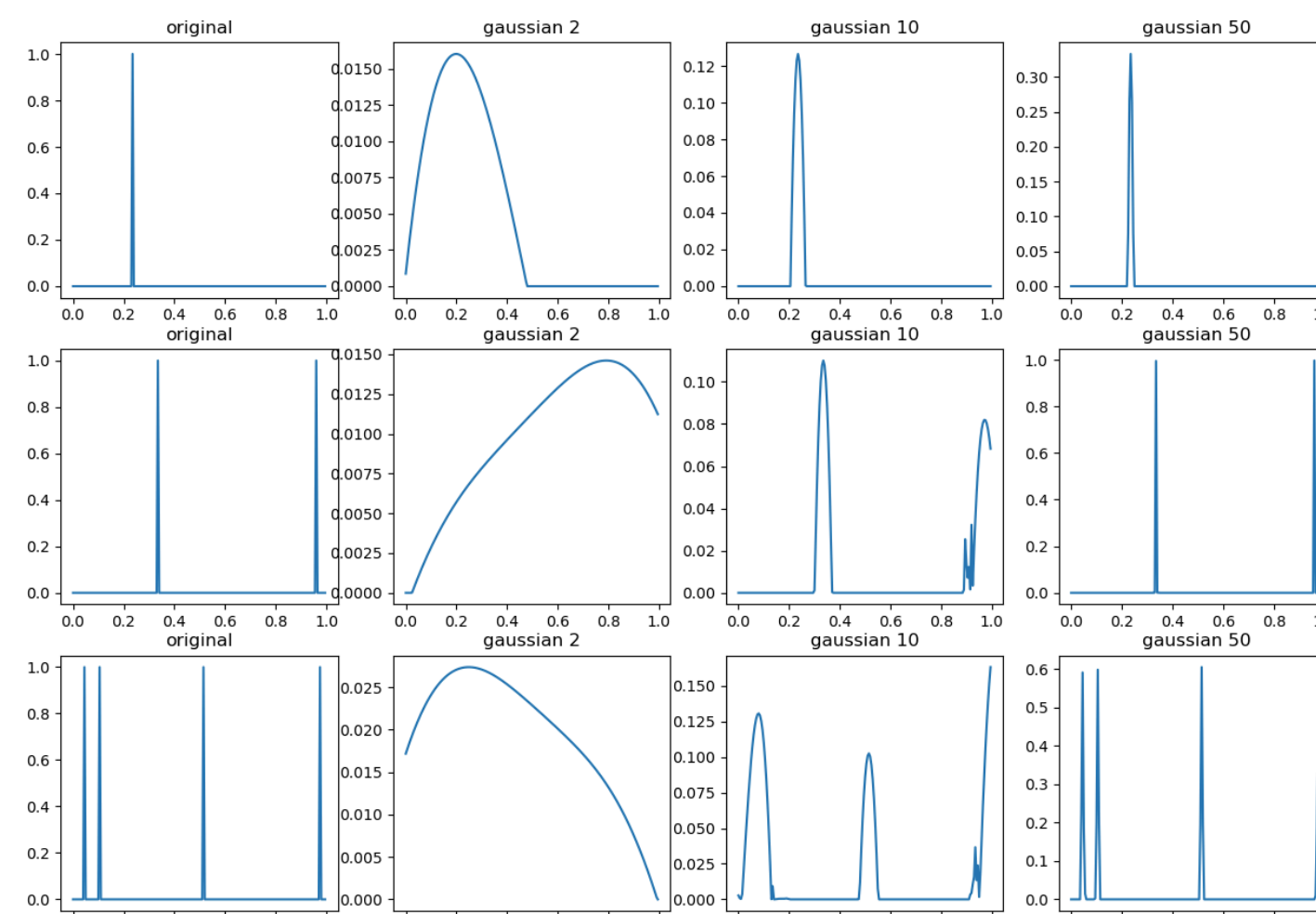
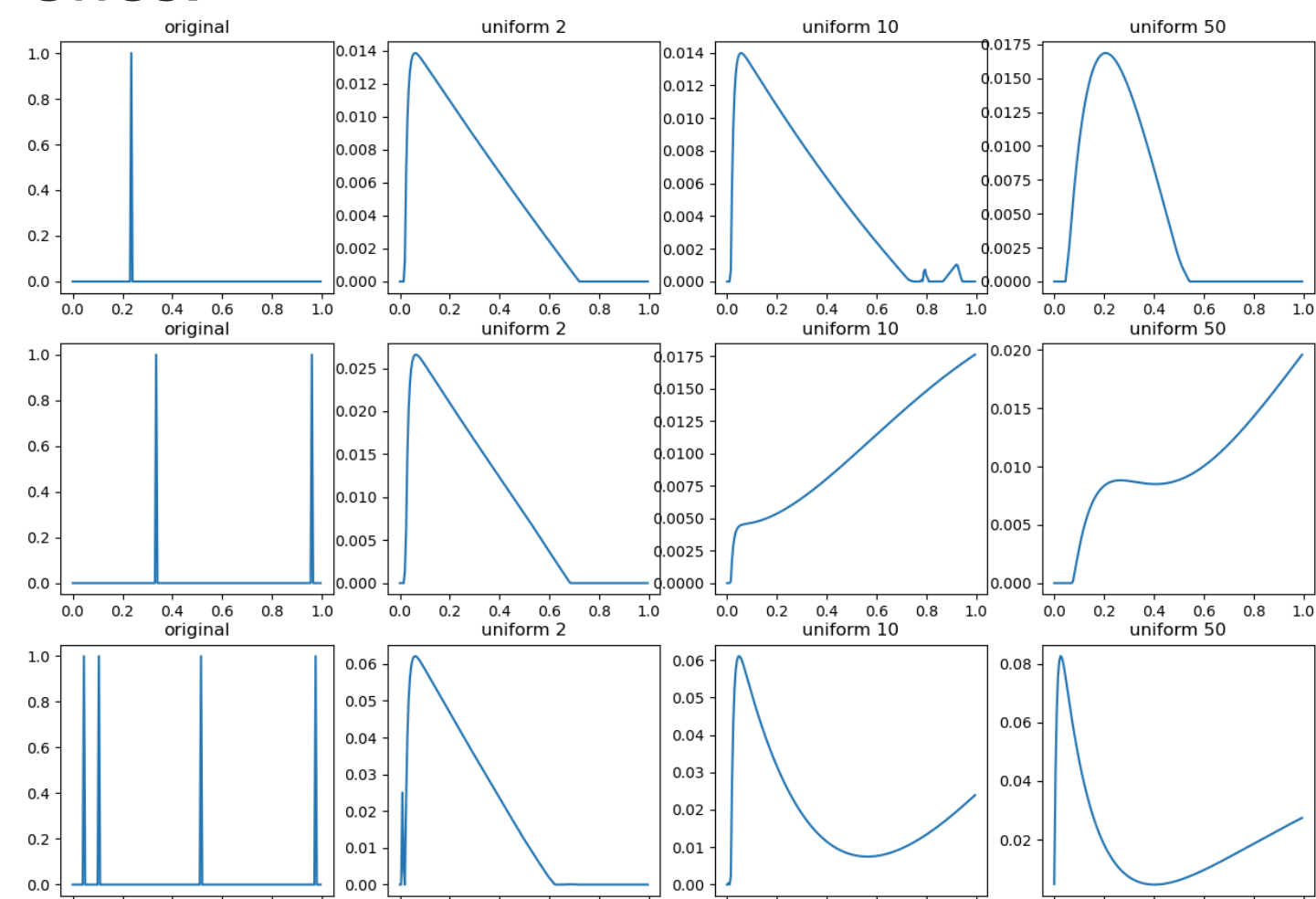


Figure 2: JESA as a commutative square; the hatch is the stop-gradient.

Kernels and reconstruction

With a general kernel k the return is $G_k = \int_0^\infty k(t) r(t) dt$, and the usual discounted return is just the exponential choice $k(t) = e^{-t/\tau}$. **Gaussian kernels reconstruct the event train markedly better than exponential or uniform ones.**



Introduction

- Planning over long horizons is intractable unless agents can break time into semantically meaningful segments.
- Frame-by-frame simulation of every feature of the world is unnecessary and prone to error accumulation, motivating temporal abstraction.

Why joint embedding?

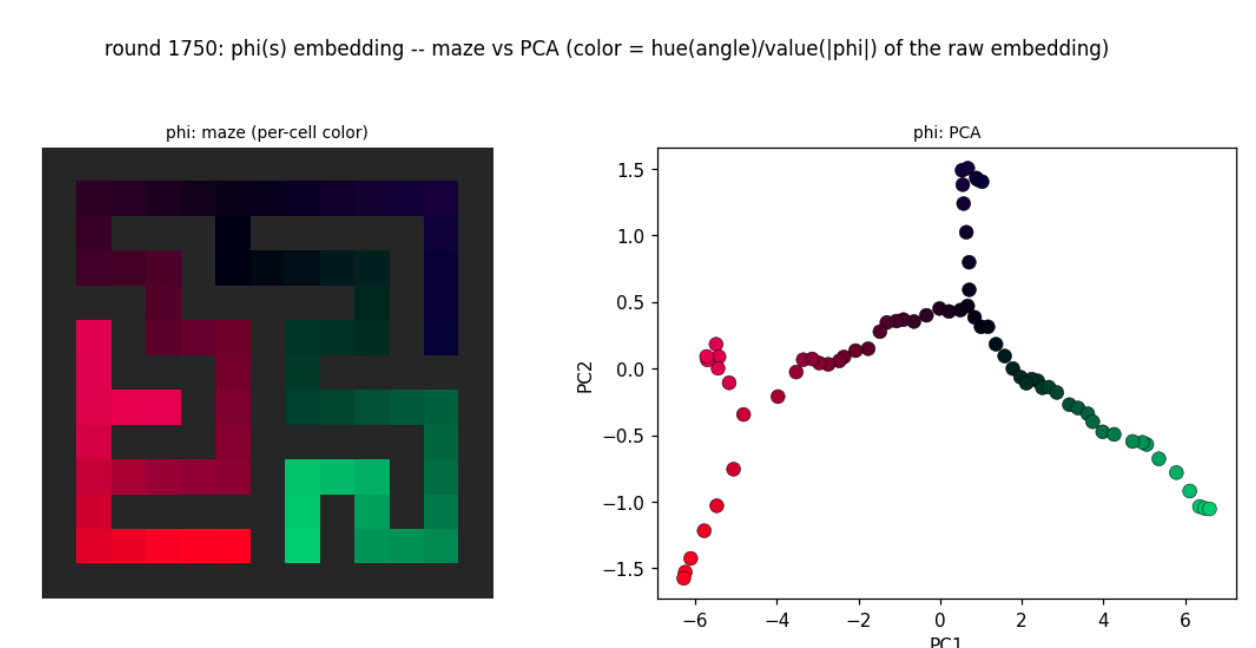


Figure 3: $\phi(s)$ over a maze (hue = angle, value = $|\phi|$) and its PCA.

ϕ comes from **contrastive successor features** (CSF), the MISL objective of Zheng et al. [2], which maximises

$$l(s; z) = \frac{H(z)}{\text{skill diversity}} - \frac{H(z | s)}{\text{identifiability}},$$

ordering states by reachability, not appearance.

Successor features $\psi(s, z)$

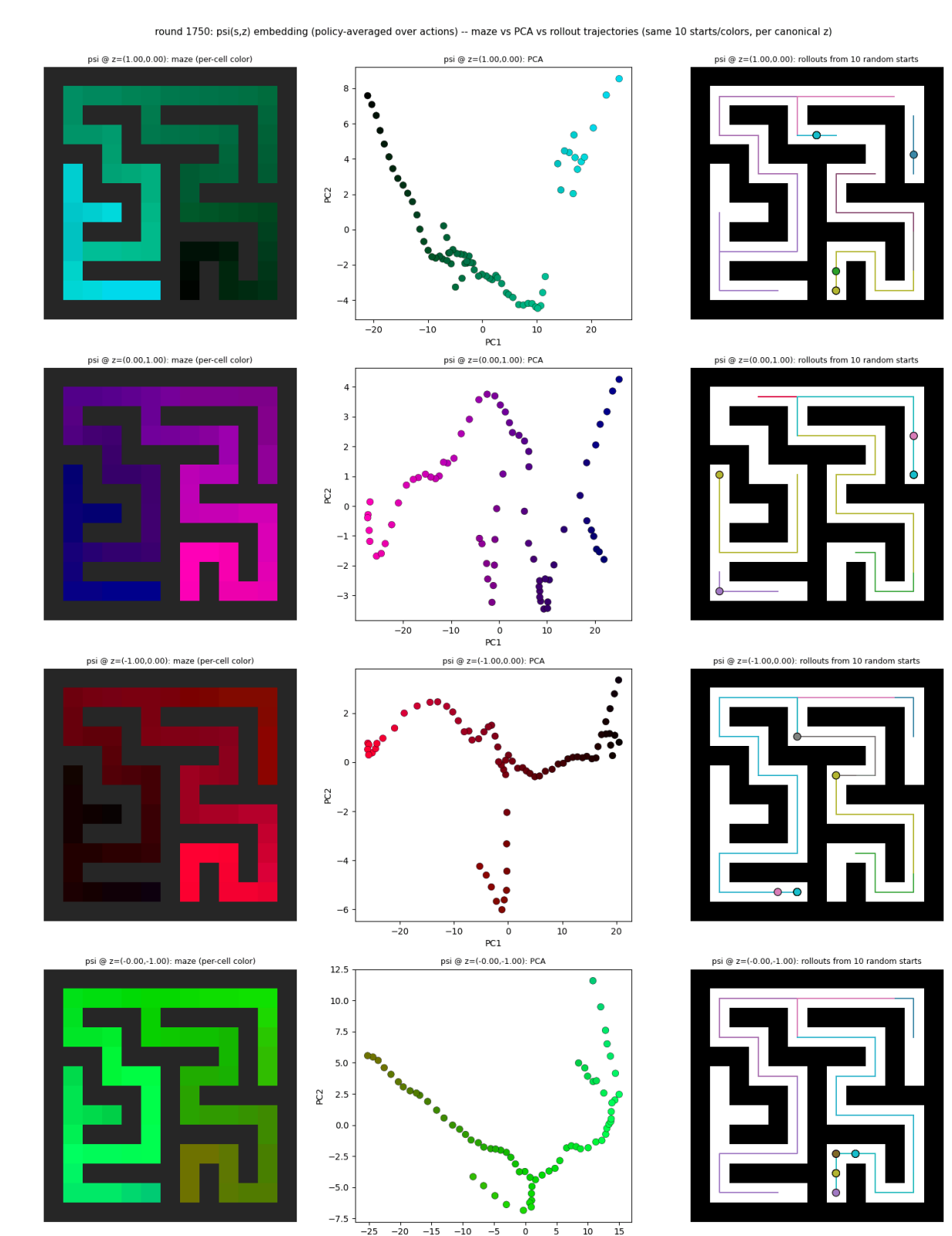


Figure 4: $\psi(s, z)$ per canonical z : maze, PCA, roll-outs.

$\psi(s, z) = \mathbb{E}[\sum_k \gamma^k \phi(s_k) | s_0 = s, z]$: each z traces its own long-horizon path.

References

- [1] Silver, D., et al. (2017). The predictron: End-to-end learning and planning. *ICML*.
- [2] Zheng, C., et al. (2025). Can a MISL fly? Analysis and ingredients for mutual information skill learning. *ICLR*.